Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 40: Approved Solutions

Fundamental Properties of Trees

Covers Chartrand & Zhang, Section 13.1

Status of this set

Ungraded practice. Solutions released with it.

Part I: How to Practice This Material

Four facts that solve almost everything

1. A tree of order n has size $n-1$. (Thm 13.6)
2. Any *two* of: connected, acyclic, $m = n-1$ — imply the third. (Thm 13.10)
3. A forest of order n with k components has size $n-k$. (Cor 13.8)
4. $n_1 = 2 + n_3 + 2n_4 + 3n_5 + \cdots + (k-2)n_k$. (Thm 13.12)

Two ways to count leaves

Use Theorem 13.12 directly, or set up the degree sum: with order n and size $n-1$, the degrees total $2(n-1)$. Both work; the second is a useful check.

Three mistakes to expect

1. **Assuming $m = n-1$ alone makes a tree.** It needs a second property.
2. **Counting degree-2 vertices in the leaf formula.** They contribute zero.
3. **Forgetting that a forest's size is $n-k$, not $n-1$.**

Part II: Practice Problems

Section A — Sizes and counts

1. Answer each with a number and a one-line reason.

(a) A tree has order 31. What is its size?

(b) A forest has order 20 and 6 components. What is its size?

(c) A connected graph has order 14 and size 22. How many edges must be removed to leave a spanning tree?

Approved solution

(a) 30, by Theorem 13.6.
(b) $20 - 6 = 14$, by Corollary 13.8. Each component is a tree, contributing one fewer edge than it has vertices.
(c) A spanning tree has $14 - 1 = 13$ edges, so $22 - 13 = 9$ must be removed.

2. A tree has three vertices of degree 2, five vertices of degree 4, and no other vertices of degree 2 or more. Find the number of leaves, the order, and the size.

Approved solution

By Theorem 13.12, degree-2 vertices contribute nothing:

$$n_1 = 2 + 2n_4 = 2 + 2(5) = 12.$$

Order: $12 + 3 + 5 = 20$. Size: 19.
Check with the degree sum: $12(1) + 3(2) + 5(4) = 12 + 6 + 20 = 38 = 2(19)$. ✓

Always run that check. It costs one line and catches arithmetic errors in the leaf formula.

Section B — Existence

3. Decide whether each tree exists. Justify.

(a) A tree of order 100 with every vertex of degree 1 or 3.

(b) A tree of order 100 with every vertex of degree 1 or 5.

Approved solution

(a) Yes. With n_1 leaves and n_3 vertices of degree 3, Theorem 13.12 gives $n_1 = 2 + n_3$, and the order gives $n_1 + n_3 = 100$. Substituting, $2 + 2n_3 = 100$, so $n_3 = 49$ and $n_1 = 51$.
Check: $51(1) + 49(3) = 198 = 2(99)$. ✓
(b) No. Here $n_1 = 2 + 3n_5$ and $n_1 + n_5 = 100$, so $2 + 4n_5 = 100$ and $n_5 = 24.5$ — not an integer. No such tree exists.

The method is identical; only the arithmetic differs. Set up two equations — the leaf formula and the order — and solve. A non-integer answer is a proof of impossibility.

Section C — Theorem 13.10

4. For each graph on $V = \{1, 2, 3, 4, 5\}$, say which of the three properties (connected, acyclic, $m = n - 1$) hold, and whether it is a tree.

(a) $E = \{12, 23, 34, 45\}$

(b) $E = \{12, 23, 31, 45\}$

(c) $E = \{12, 13, 14\}$

Approved solution

(a) Connected ✓, acyclic ✓, $m = 4 = n - 1$ ✓. **A tree** — it is P_5.
(b) Not connected ($\{1, 2, 3\}$ and $\{4, 5\}$), not acyclic (triangle), but $m = 4 = n - 1$ ✓. **Not a tree.**
(c) Connected? No — vertex 5 is isolated. Acyclic ✓. $m = 3 \neq 4$. **Not a tree**; it is a star plus an isolated vertex, so a forest with two components.

(b) is the point of the problem. It has exactly one of the three properties, and Theorem 13.10 requires two. Having $n - 1$ edges by itself proves nothing.

Section D — Writing

5. In five to eight complete sentences, explain why the proof that every tree of order n has size $n - 1$ needs Theorem 13.5 first. Describe where in the induction the leaf is used.

Approved solution

Model answer. The proof is by induction on the order, and the inductive step takes a tree T of order $k+1$ and needs to produce a tree of order k to which the hypothesis applies. The way to shrink a tree is to delete a vertex, but deleting an arbitrary vertex can disconnect it and so may not leave a tree at all. Deleting a *leaf* is safe: a vertex of degree 1 lies on no path between other vertices, so removing it and its single edge leaves a connected acyclic graph. Theorem 13.5 guarantees such a leaf exists in any nontrivial tree, which is exactly what the step needs. Without it the induction would have no legitimate way to reduce the order, and the argument would stall at its first move.

Full credit requires saying why an *arbitrary* vertex will not do. An answer that only describes the induction earns about half.

Section E — Challenge (optional)

6. Prove that if a tree has exactly two leaves, then it is a path.

Approved solution

Let T be a tree of order n with exactly two leaves, so $n_1 = 2$. By Theorem 13.12,

$$2 = 2 + n_3 + 2n_4 + 3n_5 + \cdots,$$

so $n_3 + 2n_4 + 3n_5 + \cdots = 0$. Every term is nonnegative, so each is zero: T has *no* vertex of degree 3 or more.
So every vertex has degree 1 or 2, with exactly two of degree 1. A connected graph whose degrees are all 1 or 2 is a path or a cycle, and a tree has no cycle. Hence T is a path. ■

The move worth keeping: a sum of nonnegative terms equalling zero forces every term to be zero. Theorem 13.12 turns a statement about leaves into exactly that.

Instructor notes

- **Problem 4(b)** is the item to put on the board: one property of three, and not a tree. It is the clearest demonstration of why Theorem 13.10 says *two*.
- **Problem 3(b)** shows a non-integer answer proving impossibility — the same move as the odd-degree-sum test from Week 14.
- **Problem 5** explains why Theorem 13.5 must come first. Students who see the dependency understand how the chapter is built.
- **Problem 6's nonnegative-sum argument** is worth naming.
- Reserved for the final: Section 13.1 even-numbered exercises are good extra practice.